

HackOut'26 – Problem Statements & Guidelines

Themes: Renewable Energy Intelligence & Circular Carbon Ecosystem

HackOut'26 – General Guidelines

1. Only one problem statement must be chosen per team.

Each team is required to select a single problem statement from the provided list and work exclusively on it throughout the hackathon.

2. Technology suggestions are advisory, not mandatory.

The “Tech that can be used” section under each problem statement is provided for guidance only. Teams are free to use any other technologies, frameworks, or tools as long as they effectively address the problem statement.

3. Alignment with Impact and User Requirements.

Submissions will be evaluated on how well the solution meets the user needs and achieves the impact goals outlined in the problem statement.

4. One theme per team.

Teams must select their single chosen problem statement from either the Renewable Energy Intelligence or the Circular Carbon Ecosystem theme and are not required to address both themes.

AI-Powered Hyper-Personalized Banking for Bharat

Background:

Indian banks have made massive strides in digital infrastructure — UPI, net banking, mobile apps, video KYC — yet a large section of customers (especially in Tier 2/3/4 towns and rural India) still find banking apps confusing, generic, and disconnected from their actual financial needs. At the same time, banks struggle with rising customer acquisition costs, high drop-offs in loan/product journeys, and difficulty cross-selling the right product to the right customer at the right time. Much of this stems from banks treating all customers with a one-size-fits-all digital experience, despite having rich transactional and behavioral data.

Challenge:

Design an AI-powered solution that can:

1. Analyze a customer's transaction history, spending patterns, and life-stage signals (salary credits, EMI patterns, savings behavior, etc.) to proactively recommend the most relevant banking product (loan, insurance, investment, credit card) at the right moment — instead of generic pop-up offers.
2. Simplify and personalize the digital banking journey (onboarding, loan application, KYC) using conversational AI/chatbots in vernacular languages, reducing drop-offs for first-time digital users.
3. Detect early warning signals of financial stress or fraud (unusual transaction patterns, missed EMIs, sudden behavior change) and trigger proactive, empathetic interventions rather than purely punitive actions (like immediate loan default flags).

Deliverables Expected from Participants:

- A conceptual architecture/flow diagram (data sources → AI engine → personalized output/action)
- A working prototype or wireframe showing the customer journey (e.g., a personalized dashboard, chatbot flow, or loan journey)
- An explanation of the AI/ML approach used (e.g., recommendation engines, behavioral segmentation, anomaly detection, NLP for vernacular chat)
- A note on ethical safeguards: data privacy and consent (DPDP Act, RBI data localization norms), algorithmic bias, and avoiding predatory nudging (e.g., not over-recommending loans to financially stressed customers)

Judging Criteria:

- Innovation and technical feasibility
- Depth of personalization vs. genuine customer benefit (not just upsell-driven)
- Explainability and RBI/regulatory compliance readiness
- Usability for non-tech-savvy and vernacular-first users
- Scalability across a bank's existing digital infrastructure and social/financial impact